

Earth's Core: Exploring Iron Allotropes under Extreme Conditions through First-Principles Simulations and Graph-Kernel-Based Machine Learning

Diego Juarez | IronCoreMD Project

ABSTRACT

Earth's inner core cannot be sampled directly, and experiments at simultaneous core pressures and temperatures remain difficult. We are building a first-principles and machine-learning workflow to determine how BCC, FCC, and HCP iron respond under these conditions. Quantum ESPRESSO molecular dynamics supplies energies, forces, stresses, and magnetic observables; temperature-dependent effective potential analysis supplies phonons and free energies. A Gaussian-process model based on graph-kernel structural similarity provides a route toward larger systems and longer trajectories. The current benchmark reaches an energy MAE of 7.26 meV/atom and RMSE of 9.61 meV/atom on 156 evaluation configurations. Ongoing validation emphasizes forces, phase stability, equation-of-state behavior, and magnetic sensitivity rather than energy parity alone.

SCIENTIFIC MOTIVATION

Iron dominates Earth's core, but its stable crystal structure and thermodynamic response at extreme pressure and temperature remain active questions. Small free-energy differences among BCC, FCC, and HCP iron can affect interpretations of density, elasticity, and seismic anisotropy. Magnetism adds another difficulty: local moments can persist even when long-range magnetic order is lost, so a non-magnetic calculation is not always an adequate model of high-temperature iron. Direct density functional theory (DFT) gives the needed electronic accuracy, but ab initio molecular dynamics (AIMD) is too expensive for the system sizes and time scales required to sample phase transformations or defects. Our objective is to transfer this accuracy into a machine-learning interatomic potential.

FIRST-PRINCIPLES DATA WORKFLOW

The IronCoreMD workflow begins with BCC, FCC, and HCP supercells at selected volumes and temperatures. Quantum ESPRESSO calculations produce trajectories containing atomic positions, simulation cells, total energies, forces, pressures, temperatures, and magnetic observables when available. A dedicated parser stores accepted frames in compact NPZ archives with explicit units and provenance. Temperature-dependent effective potential (TDEP) calculations provide an independent physical validation layer. Effective force constants fitted to the AIMD trajectories yield phonon dispersions, vibrational densities of states, Helmholtz free energies, and pressure-volume relations. The present repository covers BCC calculations from 4000 to 6000 K, FCC calculations from 4000 to 6500 K, and an initial HCP campaign at 5000 K. A force-ready FCC dataset at 5000 K contains 3,192 accepted 128-atom frames across eight volumes, with reproducible training, validation, and test assignments.

GRAPH-KERNEL MODEL AND HPC STRATEGY

Instead of reducing a configuration to a fixed list of handcrafted descriptors, we represent it as a graph. Atoms form nodes, local geometric relationships form edges, and labels encode the local structural environment. A graph kernel evaluates the similarity of two configurations through their related substructures. That similarity becomes the covariance used by Gaussian-process regression to learn the DFT potential-energy surface. This design is attractive for a multi-phase iron model because it compares local topology directly and can distinguish environments that change across allotropes, compression, thermal disorder, and magnetic preparation. It is also computationally demanding. Data generation requires repeated spin-polarized electronic minimization, while kernel construction involves many pairwise graph comparisons. The workflow is therefore organized for high-throughput state-point generation and accelerator-enabled kernel evaluation. Compression, provenance tracking, and deterministic dataset splits reduce avoidable data movement and make large campaigns easier to reproduce.

CURRENT RESULTS AND VALIDATION

The current graph-kernel Gaussian-process benchmark was evaluated on 156 configurations. It gives an energy mean absolute error of 7.26 meV/atom and a root-mean-square error of 9.61 meV/atom. This accuracy is encouraging, but energy agreement alone does not establish a transferable potential. Final assessment will use trajectory-grouped splits and will report force-component errors, pressure predictions, phonon stability, and free-energy differences among phases. The TDEP results already illustrate why these tests matter. BCC free-energy curves vary systematically with temperature and volume over 4000–6000 K. In an updated 4000 K comparison using 112 valid magnetic frames, collinear-magnetic and non-magnetic BCC fits are both dynamically stable. Their phonon branches differ by 0.50 THz RMS, and the magnetic fit is higher by 10.3 meV/atom. Further state points are needed before treating this difference as a phase-stability result. The next stage combines multi-phase training with forces and magnetic configurations, followed by tests of equation-of-state behavior and finite-temperature phase ordering. The resulting model will extend DFT-quality simulations to larger cells and longer time scales while retaining a transparent connection to its training calculations.

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